Justin Rodriguez

Professor Xiu

2026FA*FIN*3050J*HYJ01 (Introduction to Data Science)

Homework 2: Predicting Survival on the Titanic

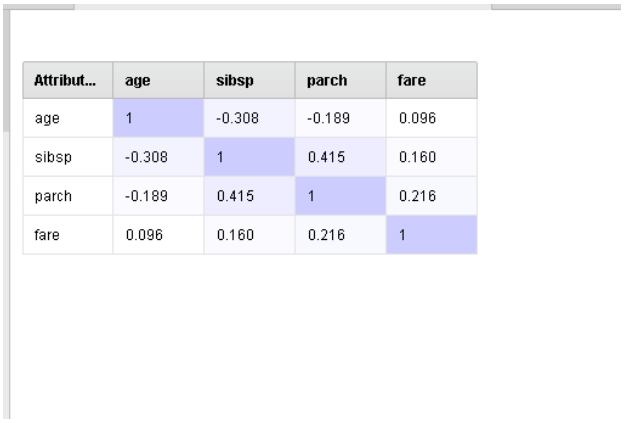
- **Descriptive Statistics:**
  - I used Altair AI Studio (RapidMiner) to explore 891 passenger records from the Seaborn Titanic dataset.

Variable	Valid n	Minimum	Maximum	Mean	Std. dev.
Age	714	0.420	80	29.699	14.526
Fare	891	0	512.329	32.204	49.693
SibSp	891	0	8	0.523	1.103
Parch	891	0	6	0.382	0.806

  - Age was missing for 177 passengers. SibSp counts siblings and spouses aboard, and Parch counts parents and children aboard.

Category	Frequency
Survived	No: 549, Yes: 342
Passenger class	First: 216, Second: 184, Third: 491
Sex	Female: 314, Male: 577
Embarked	C: 168, Q: 77, S: 644, Missing: 2

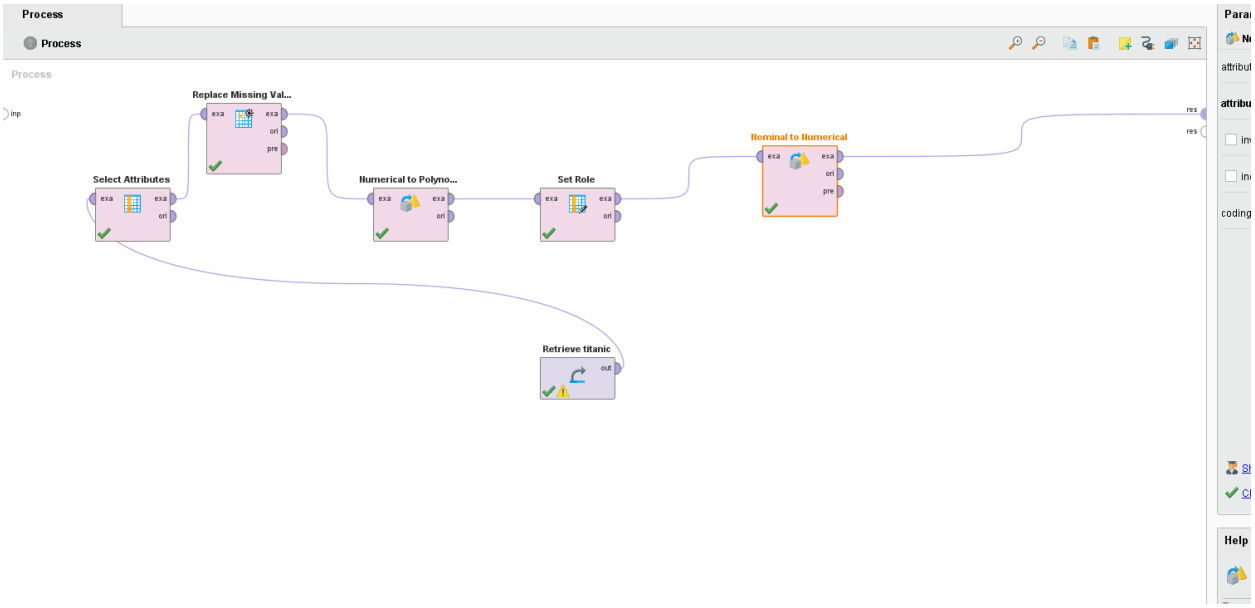
- **Pearson Correlation Matrix:**



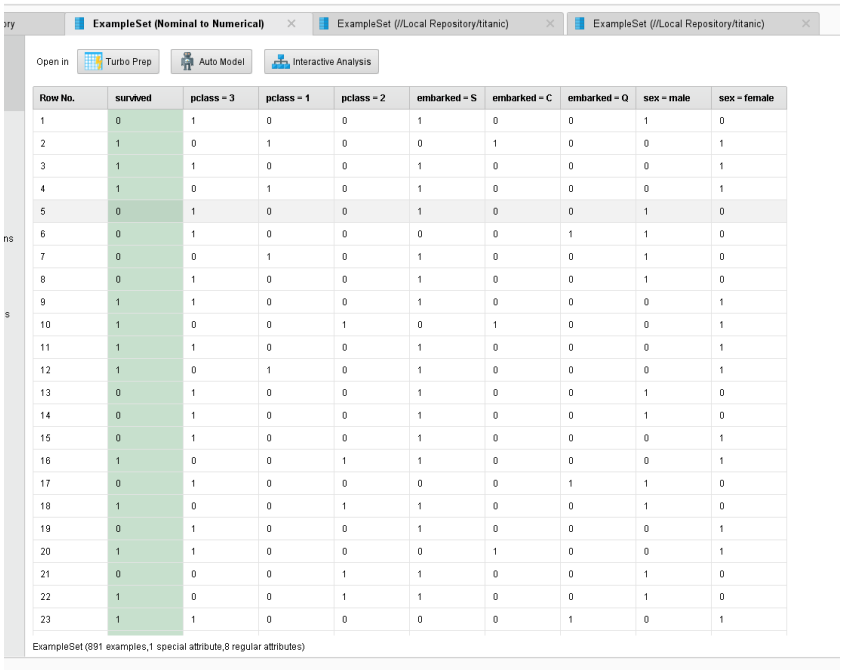
Attribut...	age	sibsp	parch	fare
age	1	-0.308	-0.189	0.096
sibsp	-0.308	1	0.415	0.160
parch	-0.189	0.415	1	0.216
fare	0.096	0.160	0.216	1

  - SibSp and Parch had the strongest correlation in the matrix (0.415). The correlation between age and fare was only 0.096.

- **Data Transformation:**
  - I kept four columns in Select Attributes: survived, pclass, sex, and embarked. The workflow included Replace Missing Values before I converted pclass and survived to categories. I assigned survived the label role and used dummy coding for the other three columns.



  - The output kept all 891 passengers and had eight indicator columns: three for class, three for embarkation, and two for sex. The first row, for example, has a 1 for third class, port S, and male.



ExampleSet (Nominal to Numerical) | ExampleSet (//Local Repository/titanic) | ExampleSet (//Local Repository/titanic)

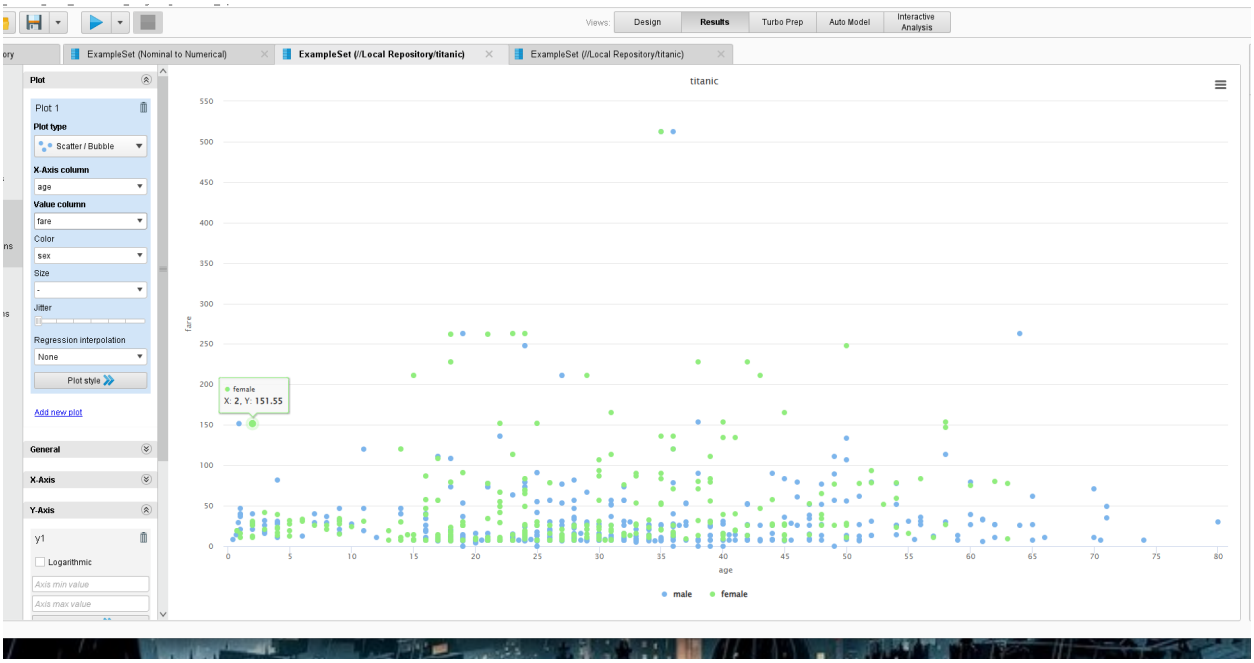
Open in: Turbo Prep | Auto Model | Interactive Analysis

Row No.	survived	pclass = 3	pclass = 1	pclass = 2	embarked = S	embarked = C	embarked = Q	sex = male	sex = female
1	0	1	0	0	1	0	0	1	0
2	1	0	1	0	0	1	0	0	1
3	1	1	0	0	1	0	0	0	1
4	1	0	1	0	1	0	0	0	1
5	0	1	0	0	1	0	0	1	0
6	0	1	0	0	0	0	1	1	0
7	0	0	1	0	1	0	0	1	0
8	0	1	0	0	1	0	0	1	0
9	1	1	0	0	1	0	0	0	1
10	1	0	0	1	0	1	0	0	1
11	1	1	0	0	1	0	0	0	1
12	1	0	1	0	1	0	0	0	1
13	0	1	0	0	1	0	0	1	0
14	0	1	0	0	1	0	0	1	0
15	0	1	0	0	1	0	0	0	1
16	1	0	0	1	1	0	0	0	1
17	0	1	0	0	0	0	1	1	0
18	1	0	0	1	1	0	0	1	0
19	0	1	0	0	1	0	0	0	1
20	1	1	0	0	0	1	0	0	1
21	0	0	0	1	1	0	0	1	0
22	1	0	0	1	1	0	0	1	0
23	1	1	0	0	0	0	1	0	1

ExampleSet (891 examples,1 special attribute,8 regular attributes)

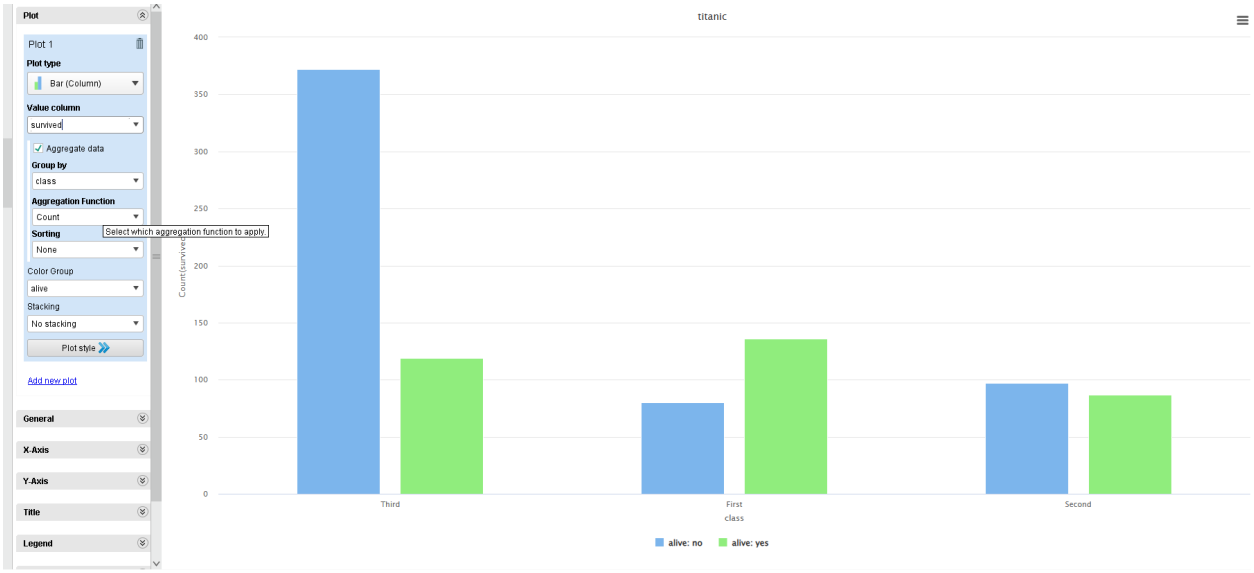
- **Visualizations:**

A. Fare versus age



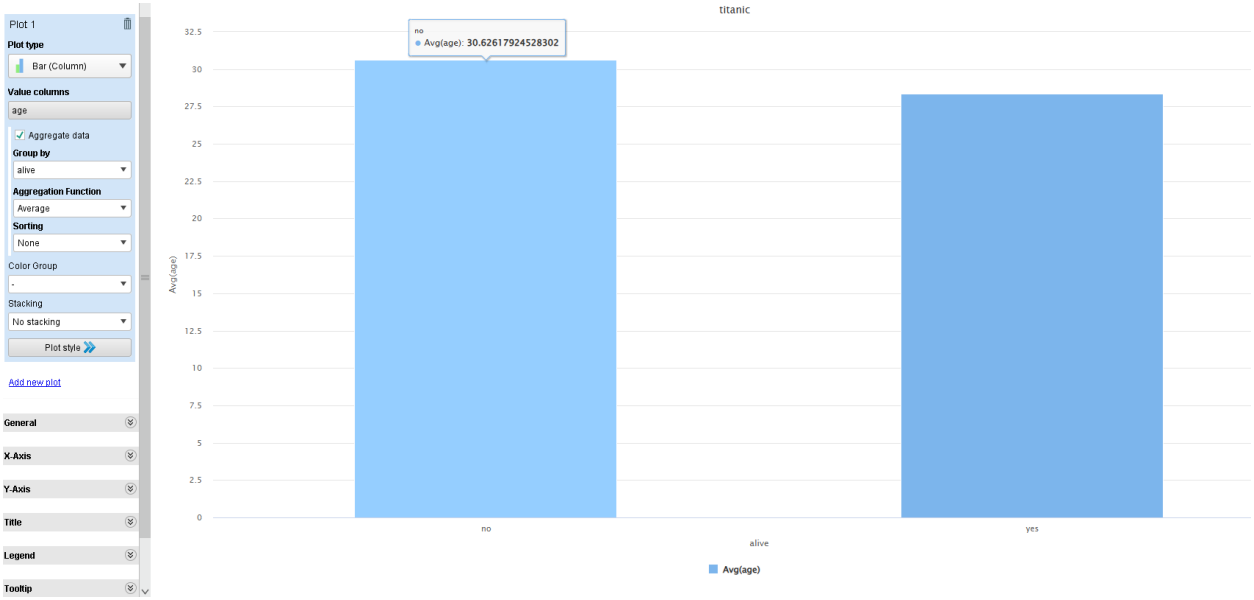
- Low fares were common across the age range. A few fares were much higher, including points above 500. The plot shows little overall relationship between age and fare.

B. Survival counts by passenger class



- Third class had the most deaths (372). First class was the only class with more survivors than non-survivors (136 versus 80). The survival rates were 63% for first class, 47% for second, and 24% for third.

C. Average age by survival status



- Among passengers with a recorded age, survivors averaged 28.34 years and non-survivors averaged 30.63 years. The difference was about 2.3 years. The averages exclude the 177 passengers with missing ages.

- **Proposed Final Project:**
  - I would like to find out how well passenger class, sex, age, fare, and family size can predict Titanic survival. The class chart got my attention because survival was much more common in first class. I want to see whether class is still useful once the model also considers age and sex.
  - I would compare logistic regression and a decision tree against a baseline that predicts the most common outcome. I would calculate family size as SibSp + Parch + 1 and use stratified cross-validation. Any missing-age replacement would be fitted using the training data in each fold. I would leave out alive because it repeats the survival outcome.
  - I would compare ROC-AUC and survivor recall and use confusion matrices to show how often each model misses a survivor.

Data source: Seaborn Titanic CSV.